Sahaj Bhadja

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Summary

Software Engineer with 2+ years of experience shipping **Java/Spring and TypeScript/React applications**, AI-enabled learning workflows, MySQL-backed services, automated tests, and observable production features.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Reduced dashboard latency from 1.2s to 350ms through **SQL indexing and query optimization**, designing normalized schemas and row-level isolation for a multi-tenant education platform supporting 500+ users reliably.
- Tripled concurrent throughput across 1K+ AI jobs by restructuring **fault-tolerant orchestration** with explicit state transitions, failure isolation, recoverable background processing, and observable workflows under sustained production load reliably.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Reduced feedback turnaround by 60% through a **production learning platform** supporting 150+ instructors and 10K+ daily requests with React, REST APIs, AWS RDS, secure ingestion, and CI/CD workflows.
- Maintained 90%+ grading consistency across 10K+ submissions through **structured model evaluation**, fallback paths, persistent results, and instructor review controls supporting reproducible automated verification and reliable learning workflows.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Improved API response times by 30% through **object-oriented microservices**, indexing, caching, pagination, and filtered payloads while troubleshooting integrations across Windows, Linux, and macOS production environments reliably at scale.
- Automated builds and 20+ unit/integration suites through **Jenkins and GitHub**, adding failure reports and cross-platform validation across Linux, Windows, and macOS with repeatable nightly workflows reliably daily.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Synchronized 2M+ records through an eight-service **React and Node.js platform** covering authentication, resources, customers, billing, leads, permissions, and notifications with WebSockets and Azure SQL across locations and workflows.
- Reduced resource wastage by 42% through **real-time data synchronization**, implementing role-based access, WebSocket updates, automated tests, Azure monitoring, and Docker/Jenkins deployments across eight modular product service domains.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Delivered **sub-200ms application APIs** across 50K+ booking records, implementing pricing, validation, payment webhooks, Firebase synchronization, and database transactions while preventing conflicting reservations across concurrent customer workflows reliably.
- Improved project delivery by 22% through a **Spring Boot API and TypeScript integration**, adding unit automation, cross-platform workflows, and maintainable service boundaries for responsive web and mobile experiences.

Skills

Languages: Java, TypeScript, JavaScript, Python, SQL, HTML, CSS
Backend & Services: Spring Boot, Node.js, Express.js, FastAPI, Flask, REST APIs, Microservices, API Design
Frontend: React, React Native, Next.js, Material UI, Tailwind CSS, Responsive Design, Reusable Components
Data & Databases: MySQL, PostgreSQL, MongoDB, Redis, AWS RDS, Azure SQL, Firebase, Data Modeling, Query Optimization
AI & Evaluation: LLMs, Prompt Engineering, Model Evaluation, Fallback Workflows, Transformers, Retrieval-Augmented Generation
Engineering Practices: Git, GitHub, Docker, Jenkins, CI/CD, Agile, Software Development Lifecycle, Unit Testing, Integration Testing, PyTest, Mockito, Observability, Production Debugging

Projects

AI-Agent Feature Delivery Harness - Design Concept

- Designed an **AI-assisted delivery harness** concept translating acceptance criteria into scoped tasks, validating generated diffs through tests and security checks, recording evaluations, surfacing guardrail failures, and requiring human review.

Learning Outcome Experiment Dashboard - Design Concept

- Designed a **React and Spring learning dashboard** concept connecting feature flags, MySQL metrics, learner outcomes, feedback, and release observations so teams can evaluate trade-offs and iterate after production deployment.

Chat2SQL | Live

- Unified **10+ database systems** in a Java and TypeScript workspace extending Chat2DB, integrating schema exploration, SQL generation, structure synchronization, and Docker-based local deployment across Windows, macOS, and Linux environments.